

Problem 2

Computing the integral for the total radiation

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In[44]:= Integrate[
  ((1 -  $\beta$  * Cos[ $\theta$ ]) ^ 2 -
    (1 -  $\beta$  ^ 2) * (Sin[ $\theta$ ]) ^ 2 *
    (Cos[ $\phi$ ]) ^ 2) /
  ((1 -  $\beta$  * Cos[ $\theta$ ]) ^ 5),
  { $\theta$ , 0, Pi}, { $\phi$ , 0, 2 * Pi}]
```

Out[44]=

$$\frac{\pi^2 \sqrt{\frac{1}{1-\beta}} (12 + 7 \beta^2)}{8 (-1 + \beta)^2 (1 + \beta)^{5/2}}$$

if +

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In[45]:= FullSimplify[%]
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$$\frac{\pi^2 \sqrt{\frac{1}{1-\beta}} (12 + 7\beta^2)}{8 (-1 + \beta)^2 (1 + \beta)^{5/2}}$$

if

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$$\begin{aligned}
 & (\beta \notin \mathbb{R} \parallel -1 \leq \operatorname{Re}[\beta] \leq 1) \&\& \\
 & (\beta \notin \mathbb{R} \parallel -1 < \operatorname{Re}[\beta] < 0 \parallel \\
 & \quad 0 < \operatorname{Re}[\beta] < 1) \&\& \\
 & (\operatorname{ArcCsc}[\beta] \notin \mathbb{R} \parallel \\
 & \quad \operatorname{Re}[\operatorname{ArcSec}[\beta]] < 0 \parallel \\
 & \quad \operatorname{Re}[\operatorname{ArcSec}[\beta]] > \pi)
 \end{aligned}$$

Below we plot the required radiation. At

first we plot what we got for this problem (0.5 and 0.2 beta values), then we will plot what we found for Bremsstrahlung.

In[47]:= $\beta = 0.5$

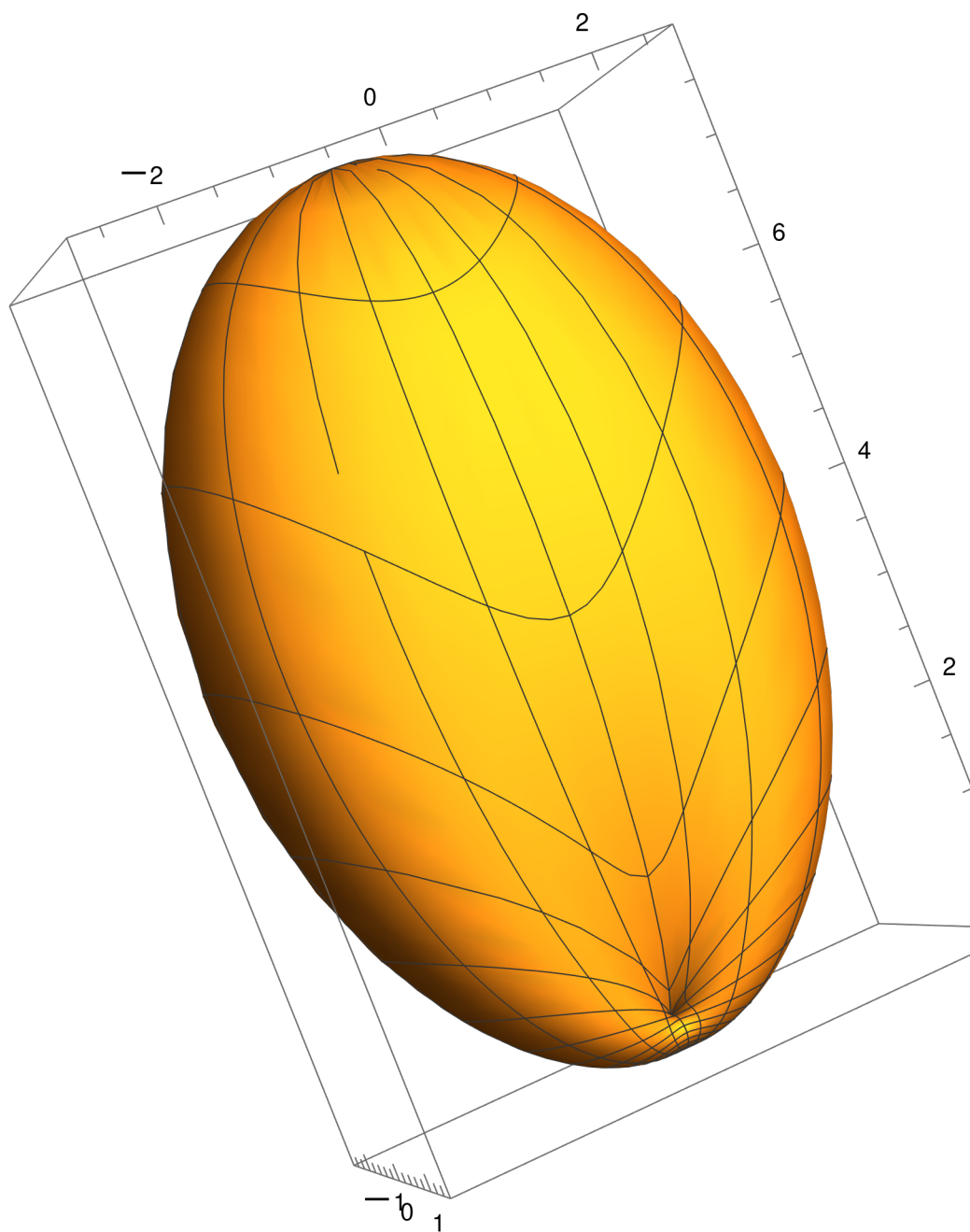
SphericalPlot3D[

$$\frac{((1 - \beta \cos[\theta])^2 - (1 - \beta^2) \sin^2[\theta] \cos^2[\phi])}{((1 - \beta \cos[\theta])^5)},$$

 $\{\theta, 0, \text{Pi}\}, \{\phi, 0, 2 * \text{Pi}\}]$

Out[47]= 0.5

Out[48]=



In[51]:= $\beta = 0.2$

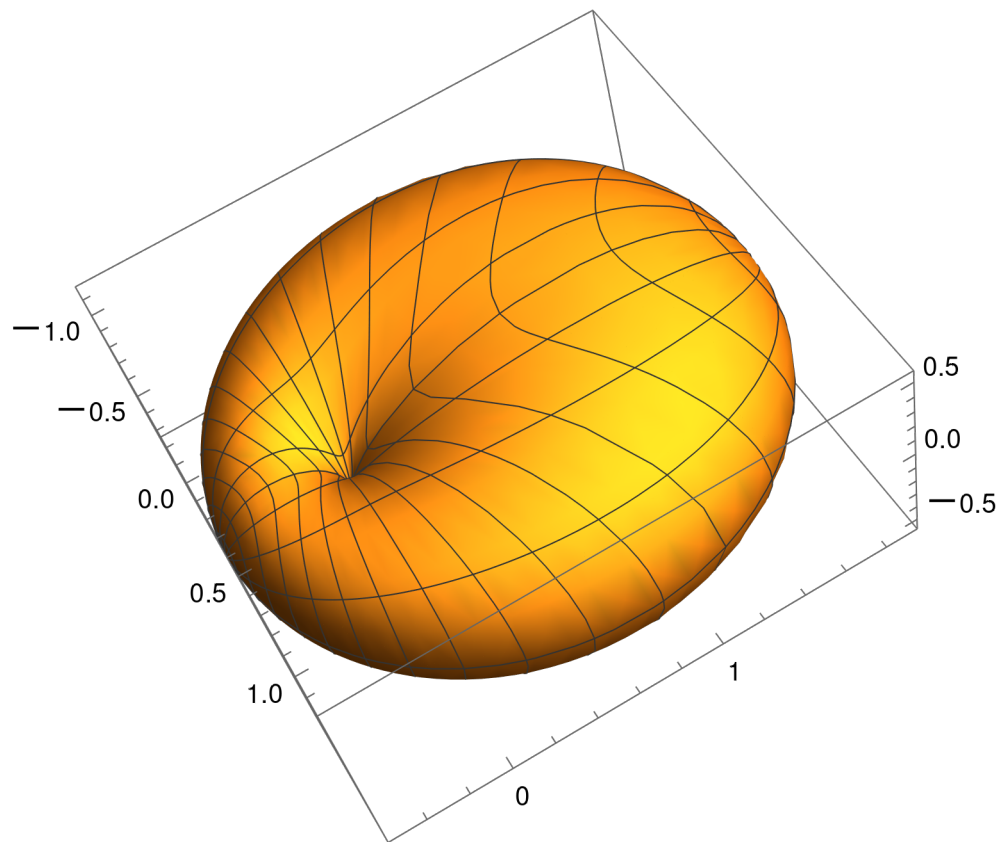
SphericalPlot3D[

$$\frac{((1 - \beta \cos[\theta])^2 - (1 - \beta^2) \sin^2[\theta] \cos^2[\phi])}{((1 - \beta \cos[\theta])^5)},$$

 $\{\theta, 0, \text{Pi}\}, \{\phi, 0, 2 * \text{Pi}\}]$

Out[51]= 0.2

Out[52]=



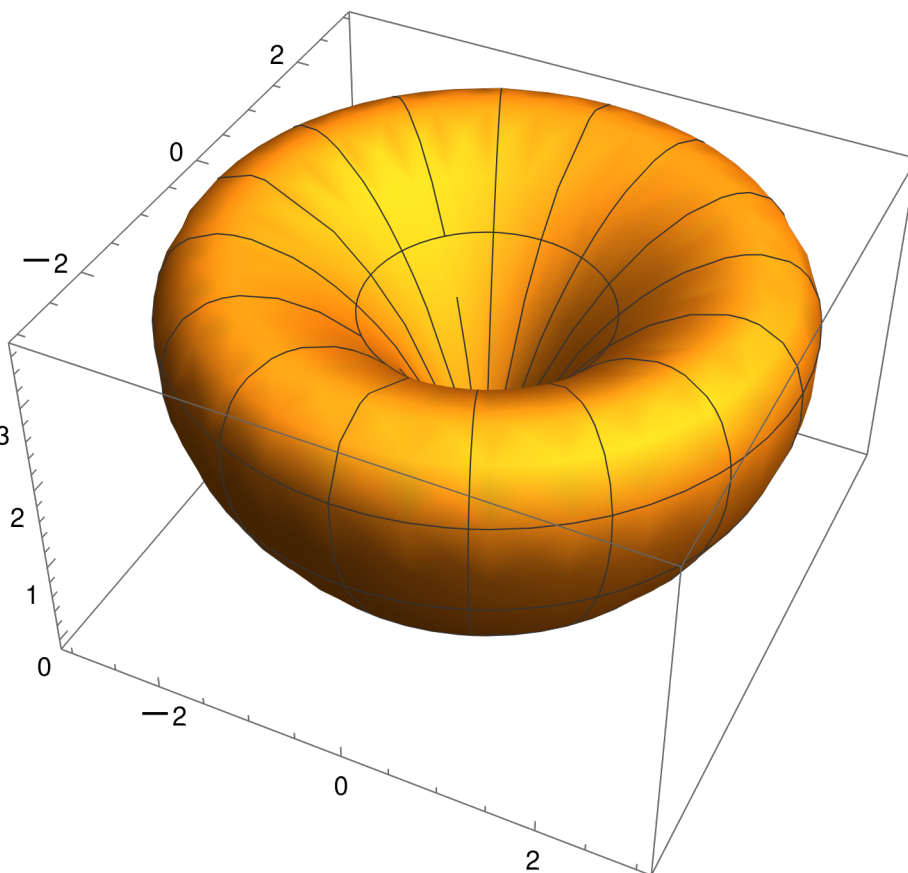
For the Bremstrahlung (beta 0.5 and 0.2)

In[53]:= $\beta = 0.5$

SphericalPlot3D[(Sin[θ]²)/
((1 - β *Cos[θ])⁵),
{ θ , 0, Pi}, { ϕ , 0, 2*Pi}]

Out[53]= 0.5

Out[54]=

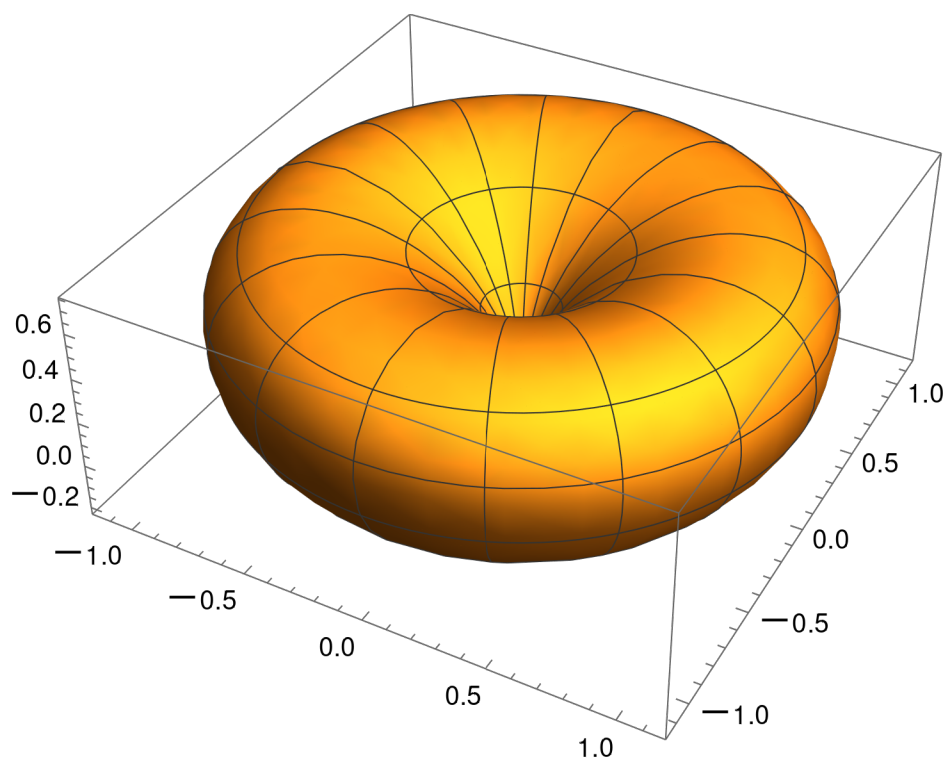


In[55]:= $\beta = 0.2$

SphericalPlot3D[(Sin[θ]²)/
((1 - β *Cos[θ])⁵),
{ θ , 0, Pi}, { ϕ , 0, 2*Pi}]

Out[55]= 0.2

Out[56]=



Problem 6 (a)

$$\text{In}[30]:= E_x' = E_x$$

$$B_x' = B_x$$

$$E_y' = \gamma * (E_y - v * B_z)$$

$$B_y' = \gamma * (B_y + (v / c^2) * E_z)$$

$$E_z' = \gamma * (E_z + v * B_y)$$

$$B_z' = \gamma * (B_z - (v / c^2) * E_y)$$

$$\gamma = 1 / \text{Sqrt}[1 - (v^2 / c^2)]$$

$$\text{Out}[30]= E_x$$

$$\text{Out}[31]= B_x$$

$$\text{Out}[32]= \frac{E_y - B_z v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Out[33]} = \frac{B_y + \frac{E_z v}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Out[34]} = \frac{E_z + B_y v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Out[35]} = \frac{B_z - \frac{E_y v}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\text{Out[36]} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Below we compute the expanded form
of $\text{vec}(E) * \text{vec}(B)$

In[39]:= $E_x' * B_x' + E_y' * B_y' + E_z' * B_z'$

$$\text{Out[39]} = B_x E_x + \frac{\left(E_z + B_y v\right) \left(B_z - \frac{E_y v}{c^2}\right)}{1 - \frac{v^2}{c^2}} +$$

$$\frac{\left(E_y - B_z v\right) \left(B_y + \frac{E_z v}{c^2}\right)}{1 - \frac{v^2}{c^2}}$$

In[38]:= FullSimplify[$E_x' * B_x' + E_y' * B_y' + E_z' * B_z'$]

$$\text{Out[38]} = B_x E_x + B_y E_y + B_z E_z$$

We verified $E \cdot B$ is invariant.

Problem 6 (b)

First let's compute $E'^2 - c^2 B'^2$

$$\begin{aligned} \text{In[41]} := & \left(E_x' * E_x' + \right. \\ & \left. E_y' * E_y' + E_z' * E_z' \right) - \\ & c^2 \left(B_x' * B_x' + B_y' * B_y' + \right. \\ & \left. B_z' * B_z' \right) \end{aligned}$$

$$\begin{aligned}
 \text{Out[41]} = & E_x^2 + \frac{(E_z + B_y v)^2}{1 - \frac{v^2}{c^2}} + \\
 & \frac{(E_y - B_z v)^2}{1 - \frac{v^2}{c^2}} - c^2 \left(B_x^2 + \right. \\
 & \left. \frac{\left(B_z - \frac{E_y v}{c^2} \right)^2}{1 - \frac{v^2}{c^2}} + \frac{\left(B_y + \frac{E_z v}{c^2} \right)^2}{1 - \frac{v^2}{c^2}} \right)
 \end{aligned}$$

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In[40]:= FullSimplify[
  (Ex' * Ex' + Ey' * Ey' +
    Ez' * Ez') -
  c^2 (Bx' * Bx' + By' * By' +
    Bz' * Bz')]
Out[40]= -((Bx^2 + By^2 + Bz^2) c^2) +
  Ex^2 + Ey^2 + Ez^2
```

Full Simplification shows that we have gotten the non-prime factors of E_i , B_j intact, that means that the spacetime-interval like quantity $E^2 - c^2 B^2$ is invariant.

Problem 6 (c)

Answer: No

Reason: For one system if $B=0$ then $S^2 = E^2 - c^2 B^2 = E^2 > 0$ which implies $S^2 > 0$

But now for another system where $E = 0$ we get $S^2 = -c^2 B^2$ which implies $S^2 < 0$

But we know that S^2 is conserved for all frames, hence it can't be greater and lower than zero for the same system in two different frames. This means the answer to the question is NO.